Teacher answers and topic guidance

Accept equivalent science in the usual response mode. Keep answers hidden during independent attempts.

Starter

1. Differentiation.
2. Two daughter cells with matching chromosome sets.
3. From higher to lower concentration overall.

Sort the case

Better movement is a possible benefit. Possible harmful effects are a risk. Recording harms is a further evidence check.

Repaired paragraph

The report describes improved movement in the group. Harm results are missing. Consider further research with safeguards. Check harms and use a suitable comparison before recommending routine use.

Fresh check

Serious harms with a small movement gain weaken support. Learners can qualify or withdraw support if they connect their judgement to the evidence.

Arrival Supported

1. The first card is Movement.
2. Movement improved in the fictional study.
3. The harm result.
4. Yes.

Arrival Independent

1. Harms.
2. Improved movement in the group studied.
3. Whether participants experienced harm.
4. Record possible harmful effects.

Arrival Aspire

1. The third card, Further check.
2. Possible improvement in movement in this group.
3. Harm data were not reported.
4. Compare movement and harms across matched groups.

Arrival Choice

1. Movement.
2. Movement improved.
3. Harms.
4. Harmful effects.

Exit Supported

1. In the group studied.
2. Possible harmful effects.

Exit Independent

1. Movement improved in the group studied. The report gives no harm results.
2. Consider further research provided it compares movement and records harms.

Exit Aspire

1. Serious harms or a similar gain in a suitable comparison group.
2. Naming the group narrows the claim to the evidence, or naming the gap explains why safety is uncertain.

Exit Choice

1. Harms are uncertain.
2. Compare benefits and harms first.

Made by Matt

Mark scheme Supported

Use the actual response and support given. These question tariffs are not participation scores. Award each distinct science point once, up to the stated maximum.

1. Replace “Everyone moves better” with a claim about the group studied.

Movement improved in the group studied.

Maximum 1 marks. Accept equivalent valid wording or a clear model demonstration.

2. Finish the sentence: The report does not tell us whether … . Explain why this matters.

It does not tell us whether harms occurred. Safety is uncertain.

Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.

3. Write or dictate one cautious recommendation with a reason.

Consider further research because movement improved. Check harms before recommending use.

Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.

Made by Matt

Mark scheme Independent

Use the actual response and support given. These question tariffs are not participation scores. Award each distinct science point once, up to the stated maximum.

1. Rewrite “It works for everyone and is safe” as two evidence-based sentences.

Movement improved in the group. The report gives no harm results, so safety is uncertain.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

2. Explain how a suitable comparison can test the benefit claim.

A suitable comparison can test whether the same improvement occurs without the approach.

Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.

3. Write a judgement that weighs both outcomes and requests a further check.

Consider further research because movement improved, provided it measures harms and compares the movement results before any decision about use.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

Made by Matt

Mark scheme Aspire

Use the actual response and support given. These question tariffs are not participation scores. Award each distinct science point once, up to the stated maximum.

1. Repair the claim that the approach caused better movement. Explain the limitation.

The group showed improved movement. A comparison is needed to test whether the approach caused it.

Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.

2. Explain how serious harms in a second study would change your judgement.

Serious harms could outweigh a small movement gain and weaken support. The judgement should change when the evidence changes.

Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.

3. Write a qualified judgement and explain which edit made it stronger.

Consider safeguarded further research only if it addresses the risk gap. Limiting the claim to this group and adding harms make the judgement fit the report.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

Made by Matt

Mark scheme Choice

Use the actual response and support given. These question tariffs are not participation scores. Award each distinct science point once, up to the stated maximum.

1. Choose a repair: “Movement improved in this group” or “Everyone benefits.” Explain.

Movement improved in this group. The evidence describes only that group.

Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.

2. Choose a repair: “Harms were not reported” or “No harms happened.” Explain.

Harms were not reported. The report cannot tell us whether any occurred.

Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.

3. Choose a repair: “Research with safeguards” or “Use now.” Explain.

Research with safeguards leaves room to record harms and compare the benefit before recommending use.

Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.

Made by Matt